

Lecture 2 — Role of Data Science in Machine Learning

Exam & Interview
Revision Sheet

Machine Learning (ML): algorithms that learn patterns from data to make predictions/decisions. **Data Science (DS):** the broader end-to-end process — collecting, cleaning, exploring, engineering, modeling, evaluating, interpreting, deploying and monitoring data-driven systems. **DS is the bridge between raw data and useful ML models.**

KEY IDEA — WHY DATA SCIENCE MATTERS

ML depends heavily on data quality. Real-world data is rarely clean — it can contain missing values, duplicates, noise, outliers, inconsistent formats and irrelevant features. A sophisticated ML algorithm cannot compensate indefinitely for poor-quality input data.

GOOD DATA + APPROPRIATE MODEL + PROPER EVALUATION → RELIABLE ML SYSTEM

Data Science Workflow



Core Building Blocks

Data Collection

Identifying & gathering data relevant to the problem — from databases, APIs, sensors, websites, surveys, transactions, apps, logs.

Example: Churn model uses age, subscription duration, complaints, spend, usage, past cancellations.

Use case: Telecoms flag customers likely to leave and act preventively.

Data Cleaning & Preprocessing

Converts raw data into usable form: handle missing values, remove duplicates, treat outliers, fix inconsistencies, encode categoricals, normalize/standardize.

Example: Missing salary imputed; Gender encoded numerically; "3 years" → 3.

Key idea: Garbage in → garbage out.

Exploratory Data Analysis (EDA)

Understanding structure & characteristics of data before modeling — distributions, trends, relationships, correlations, outliers.

Techniques: summary stats, histograms, box plots, scatter plots, correlation analysis.

Use case: Reveals house price ↑ with area; location matters; flags outlier mansions.

Feature Engineering & Selection

Engineering transforms raw data into meaningful representations; selection keeps relevant features, drops redundant/irrelevant ones.

Example: "Date of purchase" → day-of-week, month, weekend flag, days-since-last-purchase.

Exam point: Correlation with target ≠ causation.

Statistical Analysis

KEY IDEA

Statistics → understands relationships & uncertainty.

ML → uses data to learn patterns & predict.

Key concepts: probability, probability distributions, mean & variance, correlation, hypothesis testing, statistical inference, uncertainty.

Answers: Is a relationship meaningful? How much uncertainty exists? Is a pattern random? How reliable is a conclusion?

Modeling, Evaluation & Deployment

Model Development & Selection

Model choice depends on: problem type, nature of data, number/type of features, interpretability, compute constraints, desired performance.

Task Type	Example Problem
Regression	House price prediction
Classification	Spam email detection
Clustering	Customer grouping

Model families mentioned: Linear/Logistic Regression, Decision Trees, Random Forest, SVM, Neural Networks.

Model Evaluation

Classification Metrics
Accuracy, Precision, Recall, F1-score

Regression Metrics
MAE, MSE, RMSE, R^2

EXAM POINT

Techniques: **Train-test split**, **Cross-validation**. High training performance does **not** automatically mean a good model — it must generalize to unseen data.

Interpretation & Decision Making

MODEL OUTPUT → INTERPRETATION → DECISION

Example: Model predicts 90% churn probability → offer discount / better support / different plan.

Domain knowledge matters — technically correct predictions are useless if interpreted wrongly.

Deployment & Monitoring

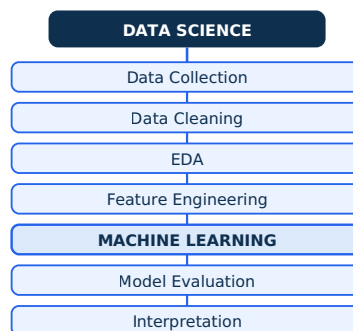
Concerns: prediction quality, **data drift**, changing user behavior, model degradation, new incoming patterns.

Example: A fraud model trained on 2025 patterns weakens if fraud strategies change in 2026.

Key idea: Deployed models need continuous monitoring & sometimes retraining.

Data Science vs Machine Learning

Data Science	Machine Learning
Broader, end-to-end lifecycle	A major component of DS
Collection, cleaning, EDA, stats, feature eng., modeling, interpretation, deployment & monitoring	Algorithms that learn patterns from data
—	Prediction, classification, clustering, decision-making
—	Requires suitable data & evaluation



Continues to: Deployment & Monitoring — closing the Data Science lifecycle loop.

Complete Example — E-commerce Customer Churn



Predict churn probability → send targeted offers / improve support → check the model still works as behavior changes.

10 THINGS TO REMEMBER

1. Data Science is broader than Machine Learning.
2. ML depends heavily on data quality.
3. Data collection must produce relevant data.
4. Preprocessing converts raw data into usable data.
5. EDA reveals patterns, relationships & anomalies.
6. Feature engineering creates useful representations.
7. Statistics quantifies relationships & uncertainty.
8. ML models must be evaluated on unseen data.
9. Predictions must become meaningful decisions.
10. Deployed models need continuous monitoring.